

# Miru Hong

PHD APPLICANT · FOOTBALL AI · MACHINE LEARNING

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Google Scholar ([https://scholar.google.com/citations?user=jsx0v\\_kAAAAJ](https://scholar.google.com/citations?user=jsx0v_kAAAAJ)) GitHub (<https://github.com/Hmiru>)

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## 01 — RESEARCH INTERESTS

Football AI, Generative Sequence Models, Counterfactual Simulation, Disentangled Representation Learning, Spatiotemporal Modeling, Sports Analytics

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## 02 — EDUCATION

University of Seoul, Dept. of Artificial Intelligence

2025–2027 (expected)

M.S. in Artificial Intelligence · Seoul, South Korea

- Advisor: Prof. Sang-Ki Ko
- Thesis (in progress): generative simulation of football from event and 3D tracking data

Inha University, Dept. of Computer Science and Engineering

2019–2025

B.S. in Computer Science and Engineering · Incheon, South Korea

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## 03 — PUBLICATIONS

\* *equal contribution* † *corresponding author*

### PEER-REVIEWED

- 01 **Miru Hong**, Minho Lee, Geonhee Jo, Hyeokje Cho, Hyunsung Kim, Pascal Bauer, Sang-Ki Ko<sup>†</sup>. "Modeling Matches as Language: A Generative Transformer Approach for Counterfactual Player Valuation in Football." *ECML-PKDD*, 2026. Applied Data Science Track. [paper] (<https://arxiv.org/abs/2603.15212>)
- 02 Minho Lee, Geonhee Jo, **Miru Hong**, Pascal Bauer, Sang-Ki Ko<sup>†</sup>. "Valuing La Pausa: Quantifying Optimal Pass Timing Beyond Speed." *MIT Sloan Sports Analytics Conference*, 2026. [paper] (<https://www.sloansportsconference.com/research-papers/valuing-la-pausa-quantifying-optimal-pass-timing-beyond-speed>)  
Research Paper Competition Finalist
- 03 **Miru Hong**\*, Minho Lee\*, Geonhee Jo, Jae-Hee So, Pascal Bauer, Sang-Ki Ko<sup>†</sup>. "ScoutGPT: Capturing Player Impact from Team Action Sequences Using GPT-Based Framework." *Hudl Conference*, 2025. [paper] (<https://arxiv.org/abs/2512.17266>)
- 04 Geonhee Jo, **Miru Hong**, Han-Jun Choi, Minho Lee, Pascal Bauer, Sang-Ki Ko<sup>†</sup>. "Imputing Multi-Agent Trajectories from Event and Snapshot Data in Soccer." *CIKM*, 2025. [paper] (<https://doi.org/10.1145/3746252.3760868>)
- 05 Minho Lee, Geonhee Jo, **Miru Hong**, Pascal Bauer, Sang-Ki Ko<sup>†</sup>. "exPress: Contextual Valuation of Individual Players within Pressing Situations in Soccer." *MIT Sloan Sports Analytics Conference*, 2025. [paper] (<https://www.sloansportsconference.com/research-papers/contextual-valuation-of-individual-players-within-pressing-situations-in-football>)
- 06 Minho Lee, Geonhee Jo, **Miru Hong**, Pascal Bauer, Sang-Ki Ko<sup>†</sup>. "Contextual Evaluation of Individual Contributions from Pressing Situations in Football." *MLSA Workshop @ ECML-PKDD*, 2025. [paper] ([https://link.springer.com/chapter/10.1007/978-3-032-15165-0\\_3](https://link.springer.com/chapter/10.1007/978-3-032-15165-0_3))

IN PREPARATION

01 **Miru Hong**, et al. Generative simulation of football from event and 3D tracking data. In preparation, 2026. Extends the event-only model with tracking data for trajectory generation, higher-fidelity simulation, and context-aware player embeddings. *Also the basis of the M.S. thesis.*

DOMESTIC CONFERENCES (KOREAN)

- 01 **Miru Hong**, Minhoo Lee, Sang-Ki Ko<sup>†</sup>. "Soccer Cross Outcome Prediction with 3D Ball Trajectories and Contextual Information." *Korea Computer Congress (KCC)*, 2026. **Outstanding Paper Award**
- 02 Geonhee Jo, **Miru Hong**, Sang-Ki Ko<sup>†</sup>. "Player Position Prediction Using Soccer Event and Snapshot Data." *Korea Computer Congress (KCC)*, 2025.

04 — RESEARCH EXPERIENCE

**CIDA Lab, University of Seoul — Graduate Research Assistant** 2025–present

Computational Intelligence and Data Analytics Laboratory · Advisor: Prof. Sang-Ki Ko

- Built *ScoutGPT*, an autoregressive Transformer that treats a football match as language: each event is decomposed into discrete attributes (actor, location, action type) and predicted by next-token prediction, conditioned on lineup and match context.
- Trained on 6.1M K League events across 2,283 matches (2021–2025); added a value-prediction head aligned to VAEP, outperforming CatBoost, LSTM and sequence-model baselines on next-event and goal prediction.
- Enabled counterfactual transfer simulation via Monte Carlo sampling — simulating a player into a new lineup and measuring the change in offensive progression and goal probability. Cut post-transfer VAEP forecast error against a naive carry-over baseline.
- Extending the framework to 3D tracking data for trajectory generation and context-aware player embeddings, with role-masking to keep positional role from acting as a shortcut in cross-season player retrieval.

**Pixelscope — Industry Collaboration** 2026, 3 months

- Built a pipeline that derives structured event data from raw player tracking data, reaching F1 > 0.90 on most kick events and F1 > 0.85 on ball control.

**K League — Data Analysis Contributor** 2025–2026

- Contributed to the 2025 K League end-of-season analytical review (Technical Study Group report).
- Taught at the K League Data Academy in 2025 and 2026, and produced its teaching material.

05 — TEACHING EXPERIENCE

**Football Data Analytics — Teaching Assistant** Fall 2025

University of Seoul · general elective

**Reinforcement Learning — Teaching Assistant** Fall 2025

University of Seoul

**Trustworthy AI — Teaching Assistant** Spring 2026

University of Seoul

06 — AWARDS & HONORS

**Research Paper Competition Finalist, MIT Sloan Sports Analytics Conference** 2026

for "Valuing La Pausa: Quantifying Optimal Pass Timing Beyond Speed"

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07 – TECHNICAL SKILLS

|               |                                                                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Programming   | Python (primary), C++                                                                                                             |
| ML / Data     | PyTorch, XGBoost, CatBoost, scikit-learn, pandas, NumPy                                                                           |
| Methods       | Autoregressive sequence models, Transformers, representation and disentangled learning, causal inference, spatiotemporal modeling |
| Football data | Event data (VERSA, VAEP), 2D/3D tracking data, event-tracking synchronisation, K League and Bundesliga datasets                   |
| Tools         | Git, Docker (multi-GPU lab cluster), Linux, LaTeX                                                                                 |
| Languages     | Korean (native), English (working proficiency), Spanish (elementary)                                                              |

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08 – TALKS & PRESENTATIONS

**"Modeling Matches as Language: A Generative Transformer Approach for Counterfactual Player Valuation in Football"** 2026

ECML-PKDD, Applied Data Science Track · oral and poster

**"Modeling Matches as Language: A Generative Transformer Approach for Counterfactual Player Valuation in Football"** 2026

NU Sports ML Workshop, Nagoya University (organiser: Prof. Keisuke Fujii) · preview of the ECML-PKDD paper

**"Imputing Player Coordinates and Velocities using Event Data in Football"** 2025

NU Sports ML Workshop, Nagoya University (organiser: Prof. Keisuke Fujii)

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09 – REFERENCES

Available upon request.